Mock Class Test 2026

Final Mark

Formative | Last attempt with a mark

85 / 90

Your instructor added question feedback to this assessment

ATTEMPT 5/5 (SUBMITTED 23/03/2026, 23:05)

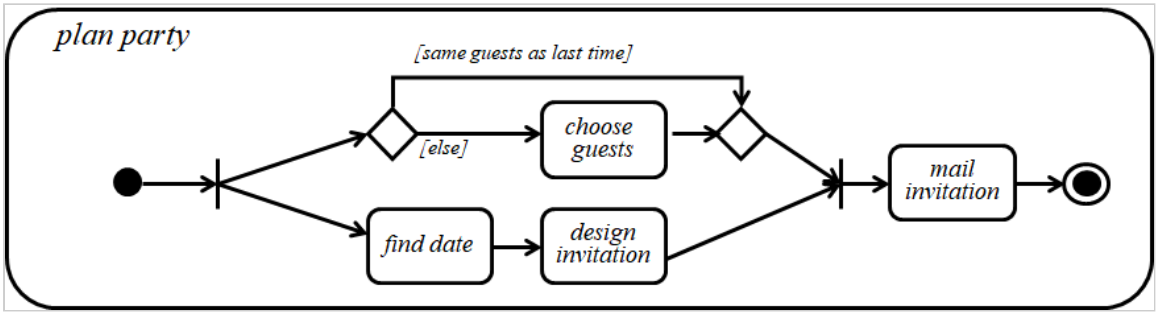
RECEIPT: 0F63D6C0D28B428290682D1739622660

85 / 90

ASSESSMENT CONTENT

1 **MULTIPLE CHOICE** **CORRECT** 2.5 / 2.5

Select all statements that are true about this Activity Diagram.



- [x] **Correct:** The action “design invitation” must appear in every execution of the activity. **50%**

Correct answer

- [ ] The action “choose guests” must always be executed before the action “mail invitation”. **-50%**
- [ ] The action “design invitation” must be executed before the action “choose guests”. **-50%**
- [x] **Correct:** The action “find date” must be executed before the action “design invitation”. **50%**

Correct answer

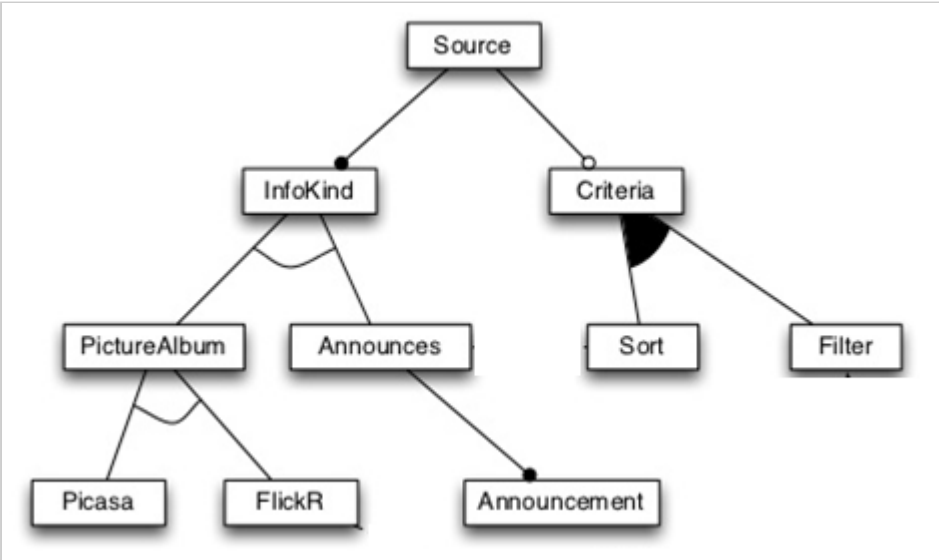
2 MATCHING CORRECT 2.5 / 2.5

Match each type of model to the most appropriate definition.

Prompts	Answers
1 Platform independent model (PIM)	**Correct:** A model with abstraction away from details about the chosen implementation technology.
2 Platform specific model (PSM)	**Correct:** A model tailored to the chosen implementation technology.
3 Descriptive model	**Correct:** A general term used for any model that describes an existing system.
4 Prescriptive model	**Correct:** A general term used for any model that describes plans for a new system.

3 MULTIPLE CHOICE CORRECT 2.5 / 2.5

Which of these are valid products of the feature diagram?



- [ ] {Source, InfoKind, PictureAlbum, Announcement} -100%

- [x] **Correct:** {Source, InfoKind, Announces, Announcement, Criteria, Sort} — 100%

Correct answer

- [ ] {Source, InfoKind, PictureAlbum, Picasa, FlickR, Criteria, Sort, Filter} — -100%
- [ ] {Source, InfoKind, Announces, Criteria, Sort, Filter} — -100%

4 MULTIPLE CHOICE CORRECT 2.5 / 2.5

Select all statements that are true about methods of classes in Class Diagrams.

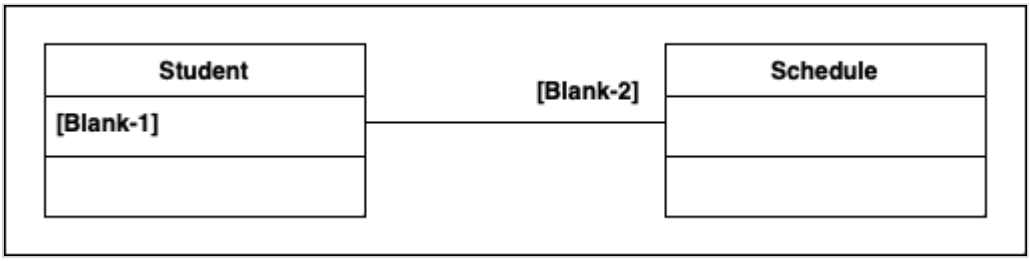
- [ ] Private methods are never shown in a Class Diagram. — -50%
- [x] **Correct:** Methods must have a name. — 50%

Correct answer

- [ ] Derived methods are executed based on other methods. — -50%
- [x] **Correct:** Methods can have a name, a type, and a visibility. — 50%

Correct answer

5 FILL IN THE BLANK CORRECT 2.5 / 2.5



Complete the class diagram above according to the following descriptions:

- ✓ **# confirmed : Boolean** The "Student" class has a protected attribute named "confirmed" of type Boolean.
- ✓ **1..10** A single "Student" object can have between 1 and 10 "Schedule" objects.

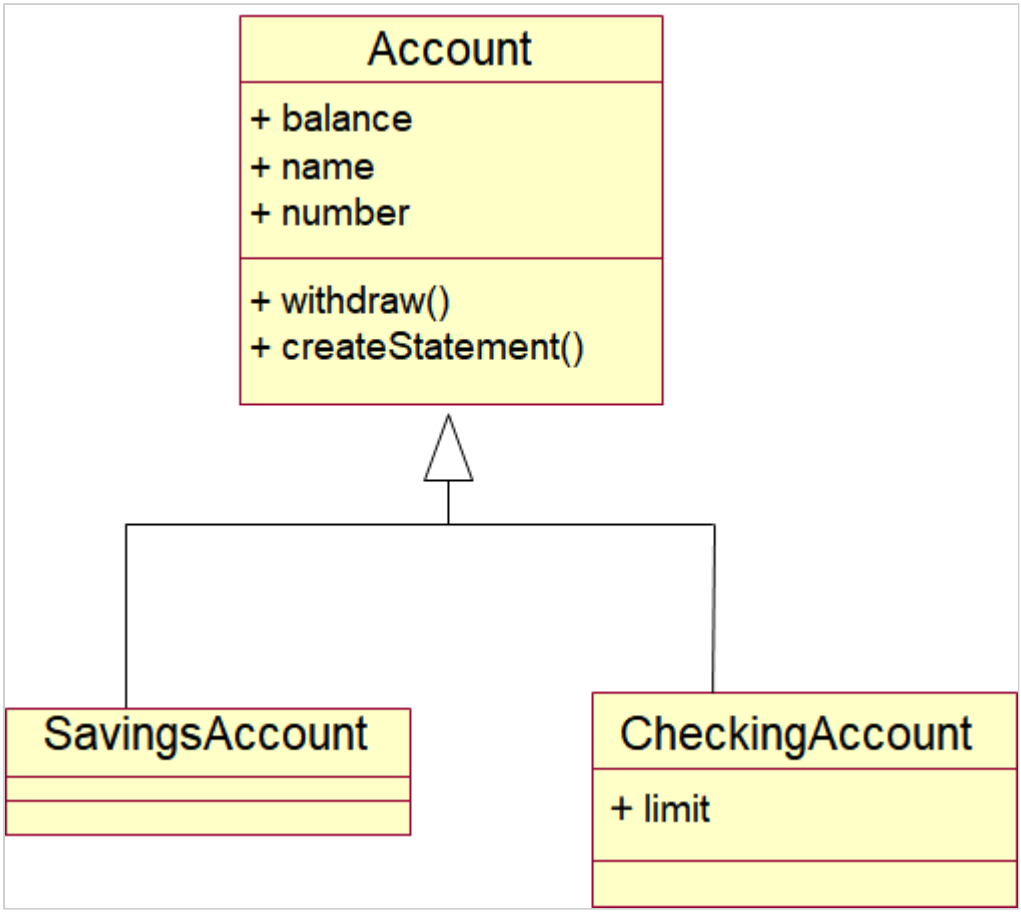
Correct

Acceptable answers for **BLANK-1** must match this pattern: (\s)*(\#)(\s)*(Confirmed|confirmed)(\s)*:(\s)*(Boolean|boolean|Bool|bool)(\s)*

Acceptable answers for **BLANK-2** must match this pattern: (\s)*(one|One|1)(\s)*(..)+(Ten|ten|10)(\s)*

6 MULTIPLE CHOICE CORRECT 2.5 / 2.5

Select all Java fragments that could be generated for this CD. Use assumptions we have made during lectures.



- [x] **Correct:** public void withdraw() {...} — 50%

Correct answer

- [ ] public Vector<String> createStatement(); — -50%
- [ ] public int balance; — -50%
- [x] **Correct:** class SavingsAccount extends Account {...} — 50%

Correct answer

7 **MULTIPLE CHOICE** CORRECT 2.5 / 2.5

Which statements are true for UseCase Specifications?

- [x] **Correct:** A UseCase specification can contain pre-conditions and post-conditions. 33.33%

Correct answer

- [ ] A UseCase specification shows all objects and their lifelines. -100%

- [x] **Correct:** A UseCase specification outlines the basic flow and alternative flows of events. 33.33%

Correct answer

- [x] **Correct:** A UseCase specification is part of a UseCase model. 33.33%

Correct answer

8 **MULTIPLE CHOICE** CORRECT 2.5 / 2.5

Select all examples that meet our definition of reactive systems.

- [x] **Correct:** Robot control software 50%

Correct answer

- [x] **Correct:** Web shop, e.g., amazon 50%

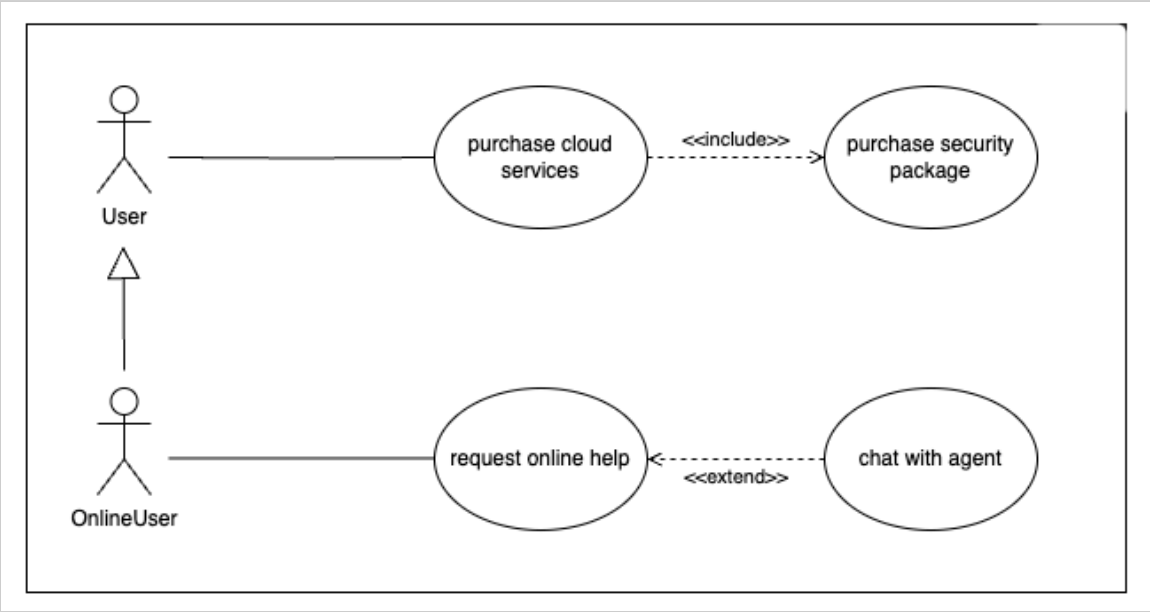
Correct answer

- [ ] sorting a list -50%

- [ ] calculation of greatest common divisor -50%

9 MULTIPLE CHOICE CORRECT 2.5 / 2.5

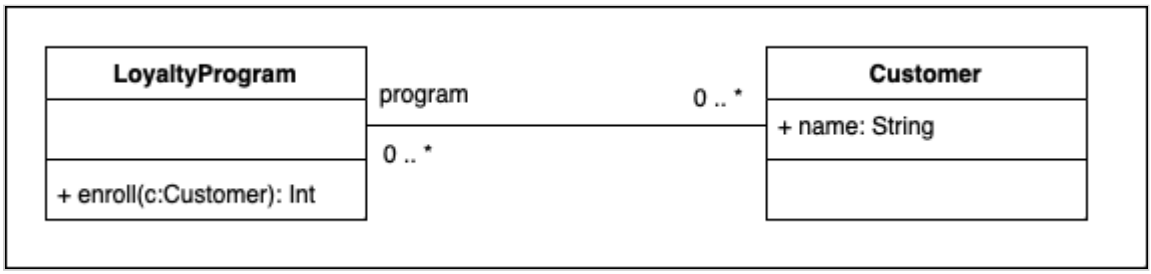
Select all statements that are true about the use case diagram below.



- [ ] "User" can initiate the use case "chat with agent". -100%
- [ ] "User" is a child of "OnlineUser" -100%
- [ ] "chat with agent" must be executed when "request online help" is executed. -100%
- [x] **Correct:** When performing "purchase cloud services", the use case "purchase security package" must be executed. 100%

Correct answer

10 FILL IN THE BLANK CORRECT 2.5 / 2.5



Consider the class diagram above. Using OCL, we will add an extra constraint - namely that every customer must be associated with 2 (and only 2) Loyalty programs. To do this, complete the following OCL invariant in it's simplest form:

✓ **context Customer**
inv: ✓ **program->size()=2**

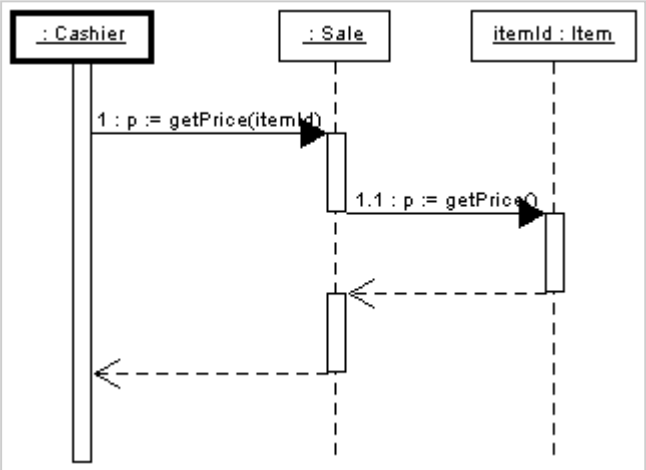
Correct

Acceptable answer for **BLANK-1** is **context customer**
Responses must match exactly

Acceptable answers for **BLANK-2** must match this pattern: (\s)*(program|Program|LoyaltyProgram|Loyaltyprogram|loyaltyProgram|loyaltyprogram)(\s)*(.|->)(\s)*(size|count)(\s)*(\((\s)*\))(\s)*(=)(\s)*(2|two|two|to)

11 MULTIPLE CHOICE CORRECT 2.5 / 2.5

What is modeled by this Sequence Diagram?



- [ ] The invocation getPrice(itemId) on the Sale object does not have a return value. **-100%**
- [ ] The order of the two messages getPrice(itemId) and getPrice() is not determined by the Sequence Diagram. **-100%**
- [ ] The Cashier object invokes getPrice() directly on the Item object. **-100%**
- [x] **Correct:** The return value of getPrice(itemId) of the Item object is the same as the return value of getPrice() on the Sale object. **100%**

Correct answer

12 MULTIPLE CHOICE CORRECT 2.5 / 2.5

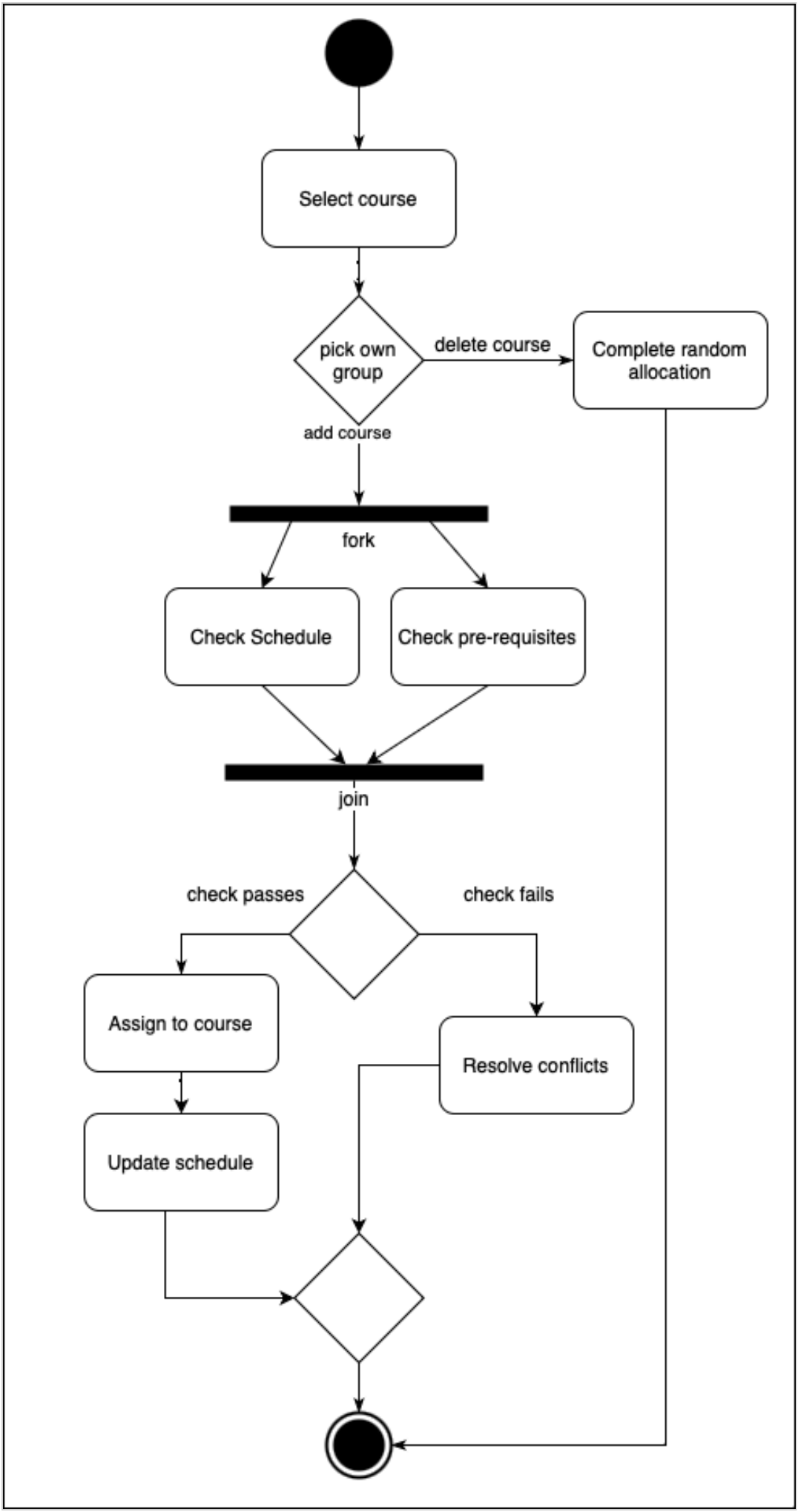
What data structures should you use to implement a to-1 association end?

- [ ] A collection. -100%
- [ ] A vector. -100%
- [x] **Correct:** An attribute of the type of the associated class. 100%

Correct answer

- [ ] An attribute of the type of the parent class. -100%

13 CALCULATED NUMERIC CORRECT 2.5 / 2.5



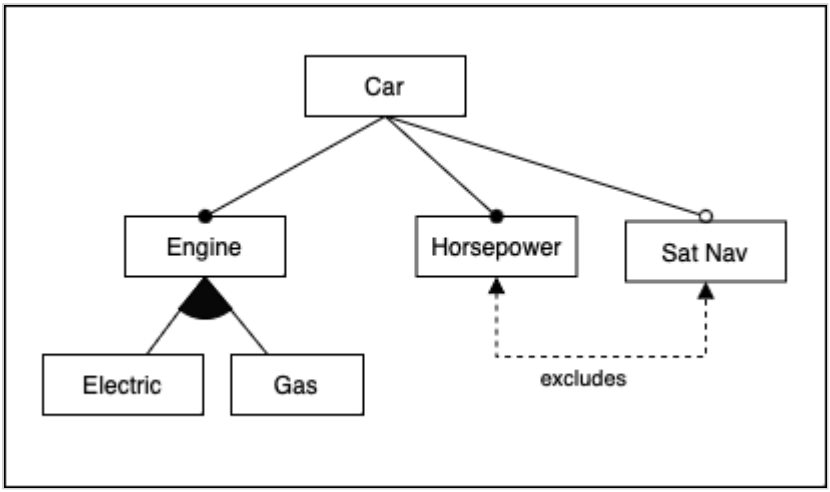
Consider the activity diagram above. Using the assumptions we used during lectures, how many possible execution paths exist?

Your answer: 5

Correct

The answer is **5**

14 FILL IN THE BLANK CORRECT 2.5 / 2.5



Consider the feature model above, and recall that features can be either core, regular, or dead. Complete the following sentences:

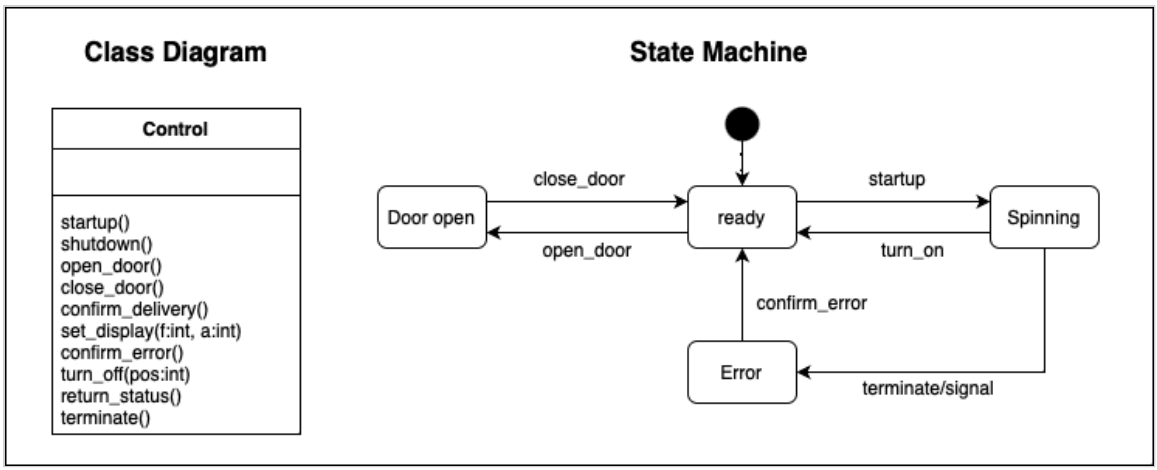
The feature "Engine" is a ✓ **core** feature. The feature "Sat Nav" is a ✓ **dead** feature.

Correct

Acceptable answer for **BLANK-1** is **core**
Responses must match exactly

Acceptable answer for **BLANK-2** is **dead**
Responses must match exactly

15 TRUE/FALSE CORRECT 2.5 / 2.5



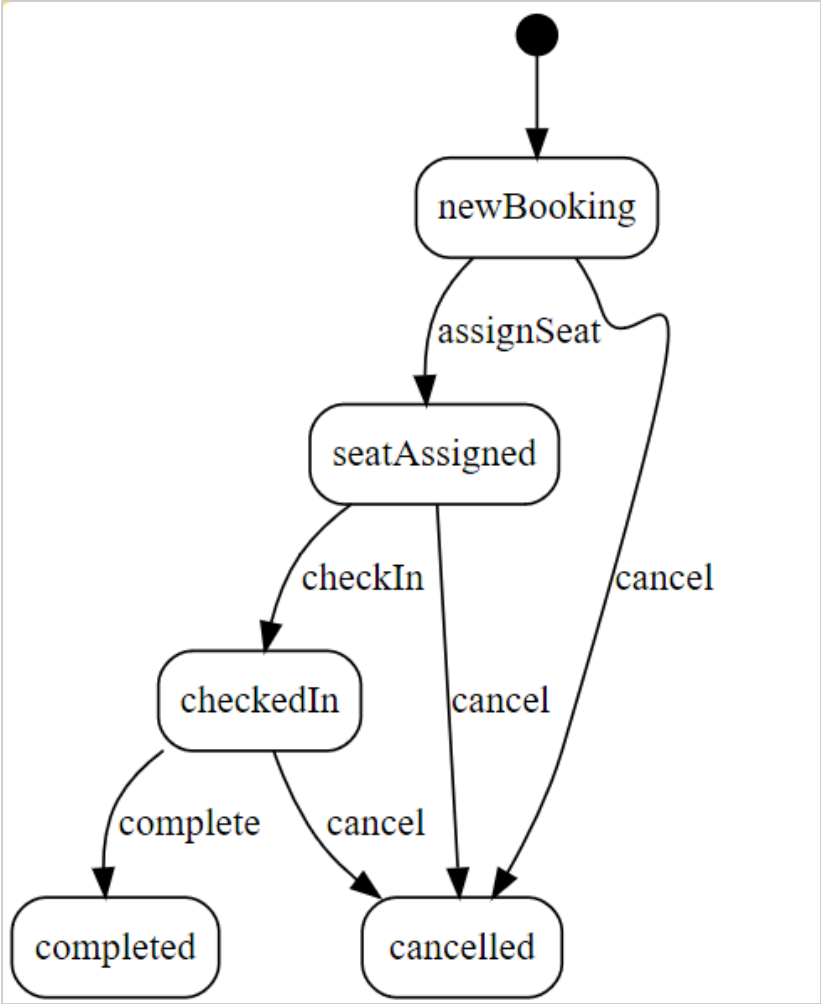
Consider the class diagram and state machine above. Is the following statement true or false:

The State Machine is consistent with the Class Diagram.

- True
- False *Correct answer*

16 FILL IN THE BLANK CORRECT 2.5 / 2.5

Fill in the gaps for this piece of code generated from the state machine:




```
public boolean complete(){
 switch(state) {
  case ✓ checkedIn
   setState( ✓ completed );
   break;
  default:
   // Other states do respond to this event
 }
}
```

Correct

Acceptable answer for **BLANK-1** is **checkedIn**
Responses must contain the acceptable answers

Acceptable answer for **BLANK-2** is **completed**
Responses must contain the acceptable answers

Feedback

Ensure you understand the rest of the code. For example, the function being called "complete" means we are modelling the event "complete" in the statement. Pick a different event, what would the transition be?

17 **MULTIPLE CHOICE** CORRECT 2.5 / 2.5

Which statements are true for states in State Machines?

- [x] **Correct:** States can have multiple outgoing transitions. 33.33%

Correct answer

- [x] **Correct:** A State Machine can be in multiple states at once. 33.33%

Correct answer

- [ ] Final state markers can have multiple outgoing transitions. -100%

- [x] **Correct:** States of a class can be defined by the attribute values of an object. 33.33%

Correct answer

18 **MULTIPLE CHOICE** CORRECT 2.5 / 2.5

Which statements are true for Object Diagrams?

- [ ] Object diagrams model sequences of messages. -50%

- [ ] Objects must have a name. -50%

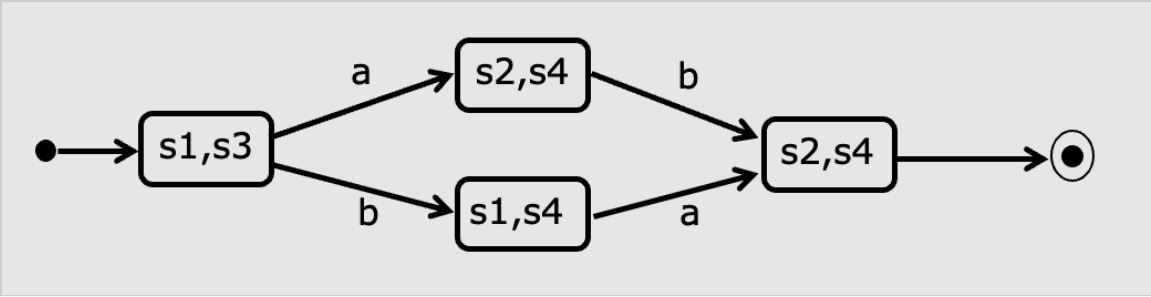
- [x] **Correct:** Object diagrams offer a static view of the system. 50%

Correct answer

- [x] **Correct:** Object names have to be underlined in object diagrams. 50%

Correct answer

19 **MATCHING** CORRECT 2.5 / 2.5



Consider the state machine above - a flattening of the State Machine shown on Slide 2 of the Mock Class Test lecture slides. Complete each statement by selecting the appropriate ending.

Prompts	Answers
1 This attempt to flatten the AND region is	**Correct:** invalid
2 This attempt to flatten the AND region does	**Correct:** not show an example of state explosion.

20 FILL IN THE BLANK CORRECT 2.5 / 2.5

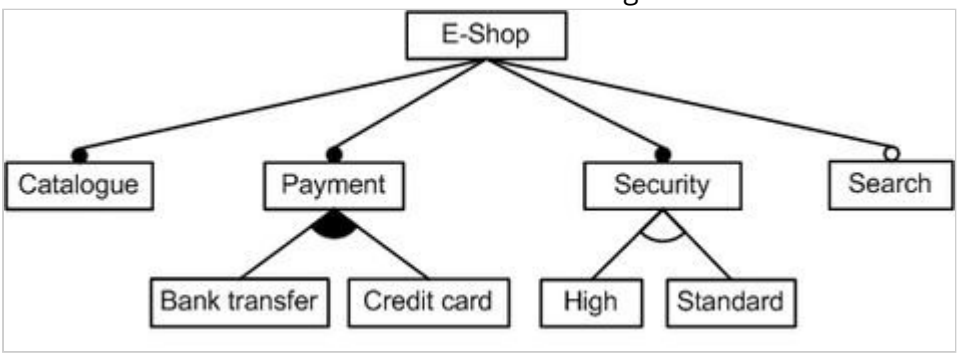
In a state machine with AND regions, a transition "a/b" is an example of ✓ **broadcasting** between regions.

Correct

Acceptable answer for **BLANK-1** is **broadcasting**
Responses must match exactly

21 MULTIPLE CHOICE CORRECT 2.5 / 2.5

Select all statements that are true about the feature diagram.



- [ ] Search is a variation point — -50%
- [x] **Correct:** Security is a variation point — 50%

Correct answer

- [x] **Correct:** Search is optional — 50%

Correct answer

- [ ] "Bank transfer" excludes "Credit card" — -50%

22 MULTIPLE CHOICE CORRECT 2.5 / 2.5

What are the types used in the Object Constraint Language (OCL)?

- [x] **Correct:** Numeric types, e.g., Integer and Real — 25%

Correct answer

- [x] **Correct:** Basic types, e.g., String and Boolean — 25%

Correct answer

- [x] **Correct:** User-defined model types from a Class Diagram — 25%

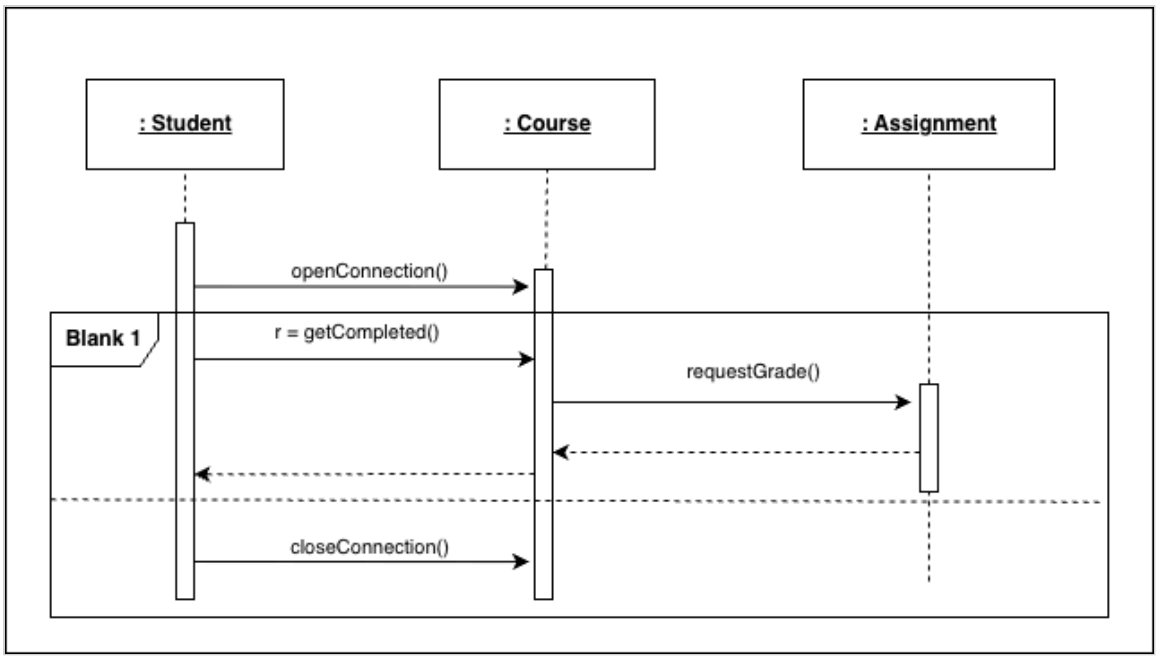
Correct answer

- [x] **Correct:** Collection types, e.g., Set, OrderedSet, and Bag — 25%

Correct answer

23 FILL IN THE BLANK INCORRECT 0 / 2.5

Complete **Alternative** in the sequence diagram below.



Incorrect

Acceptable answers for **BLANK-1** must match this pattern: (alt|Alternative|alternative

24 **MULTIPLE CHOICE** CORRECT 2.5 / 2.5

Select all statements that contain a valid transition corresponding to the fragment of Umple code defining a state machine (where the states are named "NoSchedule", "Ready", "StudentsFound" and "ScheduleCreated").

```
 ...
NoSchedule {
 begin -> Ready;
}
Ready {
 getListOfStudents -> StudentsFound;
 cancel -> NoSchedule;
}
StudentsFound {
 cancel -> Ready;
 abort -> NoSchedule;
 allocate -> ScheduleCreated;
}
ScheduleCreated {
 restart -> NoSchedule;
}
```

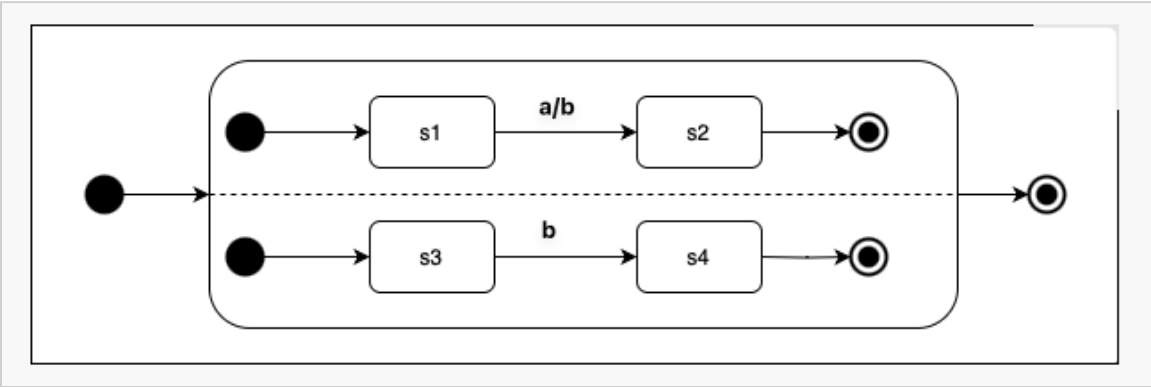
- [x] **Correct:** From "Ready" to "NoSchedule" 50%

Correct answer

- [ ] From "ScheduleCreated" to "StudentsFound" -50%
- [ ] From "NoSchedule" to "StudentsFound" -50%
- [x] **Correct:** From "NoSchedule" to "Ready" 50%

Correct answer

25 FILL IN THE BLANK CORRECT 2.5 / 2.5



Consider the state machine above. Using assumptions we have used during the lectures, complete the sentence below.

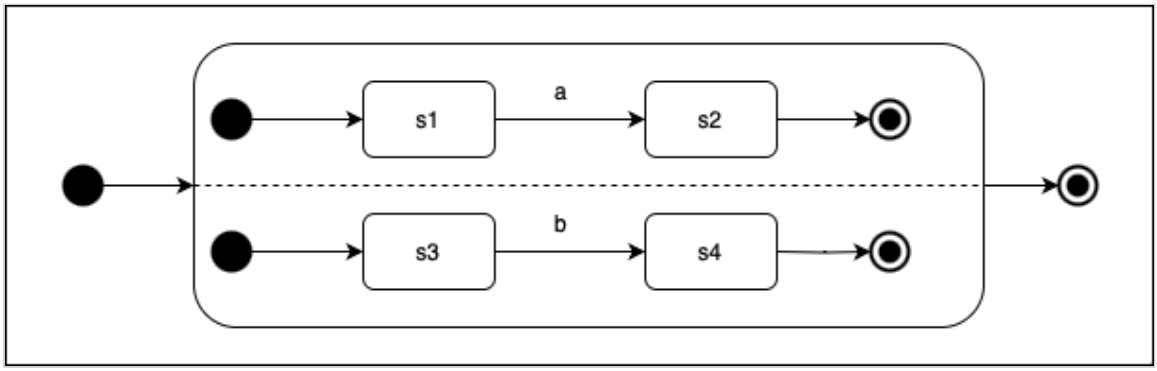
The state machine execution begins. After an event "a" occurs, the state of this state machine is (✓ s2, ✓ s4).

Correct

Acceptable answers for **BLANK-1** must match this pattern: (\s)*s(\s)*(2|Two|two|To|to)(\s)*

Acceptable answers for **BLANK-2** must match this pattern: (\s)*s(\s)*(4|four|Four)(\s)*

26 FILL IN THE BLANK CORRECT 2.5 / 2.5



Consider the state machine above. Complete the following sentence:

The state machine execution begins. The initial state of this state machine is (✓ **s1**, ✓ **s3**). After an event "a" occurs, the state of this state machine is (✓ **s2**, ✓ **s3**).

Correct

Acceptable answers for **BLANK-1** must match this pattern: (\s)*s(\s)*(1|one|One)(\s)*

Acceptable answers for **BLANK-2** must match this pattern: (\s)*s(\s)*(3|three|Three)